CareerTech

AI-Powered Job Application Tracker

MERN Stack Edition (MongoDB, Express.js, React, Node.js)

1. Project Overview

CareerTech is a personal job application management platform designed to help job seekers organize their opportunities, track application progress, and improve their CVs for specific positions using artificial intelligence.

Searching for a job often involves applying to multiple companies, tracking different application statuses, and preparing CVs for opportunities with varying requirements. Managing this information through spreadsheets, notes, or emails can become difficult, especially when applications accumulate over time.

Job seekers may also struggle to understand whether their CV effectively demonstrates the qualifications requested in a job description.

CareerTech addresses these challenges by providing a centralized personal workspace where users can save job opportunities, track their applications, manage their CV, and monitor their job search activity.

The platform integrates artificial intelligence to analyze a user's CV against a specific job description. AI identifies relevant qualifications, highlights requirements that are not clearly demonstrated, and provides practical suggestions for improving the CV.

The project aims to achieve three main objectives:

- **Organize job applications:** Help users maintain a clear record of opportunities, companies, application statuses, and important details.
- **Improve job-search visibility:** Provide a personal dashboard showing application progress and recent activity.
- **Support CV improvement with AI:** Help users understand how their existing qualifications relate to job requirements and identify opportunities to present their experience more effectively.

The final application should provide a complete workflow, from saving job opportunities and tracking applications to analyzing CVs and reviewing previous AI-generated reports.

CareerTech is a personal job-search assistant, not a recruitment platform. It does not require company accounts, recruiter interfaces, automatic job applications, or integrations with external recruitment websites.

2. Users and Roles

CareerTech supports one primary user role.

User / Job Seeker

Users are individuals who want to organize their job search and improve how they present their qualifications to employers. They should be able to:

- Create an account and manage their profile.
- Upload and maintain a main CV.
- Save job opportunities and relevant information.
- Track the status of their applications.

- View application statistics through a personal dashboard.
- Analyze their CV against job descriptions using AI.
- Upload an alternative CV for a specific analysis when necessary.
- Review previously generated AI analysis reports.
- Update or remove saved job opportunities.

Each user should have a private workspace containing their CV, saved opportunities, application records, and analysis history.

Users must not be able to access information belonging to other accounts.

The platform does not require administrators, recruiters, employers, or additional user roles.

3. Functional Requirements

3.1 Must Have: Required Features

The following features are essential for CareerTech to function as a complete personal job application tracking and AI-assisted CV analysis platform.

A. Authentication and User Management

The platform must provide secure authentication and personal account management.

- Users must be able to register using email and password.
- Users must be able to log in and log out.
- Users should be able to view and update their basic profile information.
- Each user must have access only to their own CV, applications, and AI analysis results.
- The application must protect private information and uploaded documents from unauthorized access.

After logging in, users should access their personal job-search workspace.

B. CV Management

CareerTech must allow users to maintain a main CV that can be used for AI analysis. The application should allow users to:

- Upload a CV in PDF format.
- View information about their uploaded CV.
- Replace their main CV with an updated version.
- Access their main CV from their personal workspace.
- Use their main CV when requesting an AI analysis.

Each user should maintain one main CV at a time.

When performing an AI analysis, users should also be able to upload a different PDF specifically for that analysis.

For example, a user may have a general Full Stack Developer CV but want to analyze a version tailored to a Frontend Developer position. Uploading this alternative CV must not automatically

replace the user's main CV.

The application should validate uploaded files and provide clear feedback when a document is invalid, unsupported, or cannot be read.

Previously saved AI analyses must preserve the information necessary to identify the CV version used, even if the main CV is later replaced.

A complex CV library or document version-management system is not required.

C. Job Opportunity and Application Management

CareerTech must allow users to save and organize job opportunities. Users should be able to record opportunities they are interested in, including positions they have not yet applied for. Each opportunity should contain relevant information, such as:

Information	Description
Job Title	Position offered by the company
Company Name	Organization offering the opportunity
Job Description	Description of the role and its requirements
Job Link	Optional link to the original job posting
Application Date	Date the user applied, when applicable
Status	Current stage of the opportunity or application
Personal Notes	Optional information about the opportunity

The platform should allow users to:

- Add new job opportunities.
- View their saved opportunities.
- Open an opportunity to review its details.
- Edit job information.
- Delete opportunities that are no longer relevant.
- Search opportunities by job title or company.
- Filter opportunities by application status.

The application should preserve the information entered by users without requiring connections to job boards or external recruitment platforms.

D. Application Status Tracking

Users must be able to track the progress of their job opportunities. The application should support the following statuses:

Status	Description
Saved	The user is interested in the opportunity but has not applied yet.
Applied	The user has submitted an application.

Status	Description
Interview	The user has been invited to an interview.
Offer	The company has made a job offer.
Rejected	The application was unsuccessful.

Users should be able to update application statuses as their job search progresses. For example, a user may save an interesting opportunity, apply several days later, and update its status to Interview after receiving a response from the employer.

The application should update the relevant dashboard statistics when a status changes.

Application statuses must be controlled by the user. AI analysis must not automatically change application statuses or determine whether a user should apply for a position.

E. Personal Dashboard

CareerTech must provide a dashboard showing an overview of the user's job search. The dashboard should include:

- Total number of saved opportunities and applications.
- Number of opportunities in each status.
- Recently added opportunities.
- Recent application activity.
- Quick access to the user's main CV.
- Access to the AI CV analysis functionality.

The dashboard should help users quickly understand the current state of their job search. For example, a user may see that they have saved 12 opportunities, applied to seven positions, received two interview invitations, and received one job offer.

All statistics must be calculated using actual information stored in the application. The dashboard should update when opportunities are created, modified, or deleted.

F. AI-Powered CV Analysis

CareerTech must integrate AI to help users understand how their qualifications relate to a specific job opportunity. The application should provide a dedicated CV analysis feature.

Users must be able to analyze their CV without creating or saving a job application first. The analysis workflow should allow users to:

- 1. Open the CV analysis page.
- 2. Select their main CV or upload a different PDF.
- 3. Paste the job description they want to analyze.
- 4. Request an AI analysis.
- 5. Receive a structured report.
- 6. Review the analysis and suggested improvements.
- 7. Access the report again through their analysis history.

The application must clearly indicate when an analysis is being processed and provide appropriate feedback if it fails.

The AI report must contain the following information:

1. Overall Assessment

A concise explanation of how closely the qualifications demonstrated in the CV align with the job requirements. The assessment should include an understandable category, such as: Strong Match, Good Match, Partial Match, or Low Match. The category must be supported by the analysis rather than presented as an unexplained judgment.

2. Matching Qualifications

Identify relevant skills and qualifications supported by the CV. For each important match, the AI should explain how the CV demonstrates the requirement. For example, if a job requires Next.js experience and the CV describes a project developed with Next.js, the AI should recognize that connection.

3. Unconfirmed Requirements

Identify important job requirements that are not clearly demonstrated in the CV. The AI should distinguish between:

- A requirement that appears unsupported by the provided information.
- A qualification that may exist but is not clearly mentioned or explained.

The absence of a skill from a CV must not automatically be treated as proof that the candidate does not possess it.

4. Relevant Experience

Highlight professional experience, projects, education, or certifications that are especially relevant to the selected position. The analysis should explain how the identified experience relates to the job requirements.

5. CV Improvement Suggestions

Provide practical suggestions for presenting the user's existing qualifications more effectively. Suggestions may include:

- Highlighting relevant projects more clearly.
- Improving descriptions of technical contributions.
- Clarifying responsibilities and achievements.
- Emphasizing existing qualifications related to the position.
- Improving wording or organization when appropriate.

The AI must not invent experience, qualifications, certifications, or professional achievements. The report should help users improve how they present their real experience rather than encourage misleading CV content.

G. AI Analysis History

CareerTech must preserve previous AI analysis reports. Users should be able to:

- View their analysis history.

- Open a previous report.
- Review the job description used for the analysis.
- Identify the CV version used.
- View the original generated results.
- Delete analysis records when no longer needed.

Each analysis should preserve the relevant CV content or document reference, job description, generated report, and analysis date.

Replacing the main CV or running a new analysis must not overwrite previous reports. Users must only access their own analysis history.

3.2 Should Have: Recommended Features

These features improve the usefulness of CareerTech and provide opportunities for additional technical development.

A. Explainable CV Match Score

The application could provide a numerical score between 0 and 100 representing the estimated alignment between the CV and job requirements. If implemented, the score should:

- Consider the importance of job requirements.
- Use qualifications demonstrated in the CV.
- Remain consistent with the generated analysis.
- Include a clear explanation of how the assessment was reached.

The score must not be presented as a prediction of hiring success. A numerical score is not required for the core application because an unexplained percentage can be misleading.

B. Application Follow-Up Dates

Users should be able to record relevant dates associated with their applications, such as:

- Application submission date.
- Expected follow-up date.
- Scheduled interview date.

This information should help users organize their job search and remember important upcoming actions.

C. Improved Application Filtering

The application should provide additional search and filtering options, such as:

- Company name.
- Job title.
- Application status.
- Date added.
- Application date.

Users should be able to find specific opportunities without manually reviewing their entire history.

D. Job Search Activity Statistics

The dashboard should provide additional information about application activity over time. Examples include:

- Applications submitted during the current month.
- Number of interview invitations.
- Changes in application activity between months.
- Distribution of applications by status.

Statistics should use actual user records.

3.3 Could Have: Optional Features

The following improvements offer opportunities for creativity and additional technical exploration.

- **Application Reminders:** Remind users about follow-up dates or upcoming interviews.
- **Downloadable AI Reports:** Allow users to export CV analysis results as PDF documents.
- **AI CV Tailoring Suggestions:** Suggest specific improvements to existing CV sections for a selected opportunity without inventing qualifications.
- **Application Notes History:** Allow users to record important updates or interactions with employers.
- **Job Search Goals:** Allow users to define personal application targets and monitor progress.

Students are encouraged to propose other features that improve job-search organization or CV preparation.

Additional functionality should remain focused on the needs of individual job seekers rather than expanding CareerTech into a recruitment management platform.

4. AI Integration

4.1 Purpose

The primary purpose of AI in CareerTech is to help job seekers evaluate how effectively their CV demonstrates the qualifications required for a specific job opportunity.

Job descriptions often contain multiple technical skills, experience expectations, and educational requirements. Candidates may struggle to determine which of their qualifications are most relevant or whether their CV clearly communicates their experience.

CareerTech uses AI to analyze a CV alongside a job description and generate a structured report highlighting relevant qualifications, unconfirmed requirements, and practical improvements.

The AI feature should help users prepare stronger, more relevant applications without making hiring predictions or inventing professional experience.

4.2 Expected AI Workflow

The AI functionality should follow these steps:

- **1. CV Selection:** The user selects their main CV or uploads a different CV for the analysis.
- **2. Job Description:** The user pastes the description and requirements of the desired position.
- **3. Document Processing:** The application extracts readable text from the selected PDF.
- **4. AI Request:** The backend provides the extracted CV content and job description to the selected AI model or service.
- **5. Qualification Analysis:** AI identifies relevant qualifications, matching skills, and requirements that are not clearly demonstrated.
- **6. Improvement Suggestions:** AI proposes practical changes that could improve the CV's relevance and clarity.
- **7. Response Validation:** The application validates the generated result before presenting it.
- **8. Report Presentation:** The user receives a structured, readable analysis.
- **9. History:** The analysis is saved so the user can review it later.

The analysis must remain independent of job application tracking. Users should be able to analyze an opportunity without saving it, and they should be able to track applications without using AI.

4.3 Expected AI Capabilities

The AI integration should be capable of:

- Understanding professional information extracted from CV documents.
- Identifying skills, experience, education, and relevant projects.
- Interpreting job descriptions and qualification requirements.
- Comparing CV information with job requirements.
- Highlighting demonstrated qualifications.
- Identifying requirements that lack clear supporting evidence.
- Explaining the relevance of professional experience.
- Generating practical suggestions for improving CV presentation.
- Producing structured results that the application can validate and display.
- Acknowledging insufficient or unreadable information.

For example, a user uploads a CV describing experience with React, Next.js, TypeScript, and Express.js. The user then pastes a job description for a Full Stack Developer position requiring Next.js, TypeScript, Node.js, and PostgreSQL.

The AI should recognize the demonstrated experience with Next.js and TypeScript and identify Express.js as relevant to the Node.js requirement. If PostgreSQL is not mentioned, the report should explain that database experience with PostgreSQL could not be confirmed from the CV. The AI could suggest making relevant database projects more visible if the candidate has such experience.

It must not recommend claiming PostgreSQL experience that the candidate does not possess.

(Note: the technologies named in this example describe a candidate's CV and a job posting. They illustrate the AI behavior only and are not part of the project's technology stack.)

4.4 Technical Considerations

The AI functionality must be integrated directly through the Express.js backend using an appropriate AI API or model. No separate AI microservice is required.

Students should consider the following constraints:

- CV documents may contain different layouts and formatting styles.
- Some PDFs may contain images or scanned pages that are difficult to process using ordinary text extraction.
- The application should clearly handle unreadable or unsupported documents.
- AI must not invent candidate qualifications, experience, or achievements.
- Missing CV information must not automatically be interpreted as a missing skill.
- Generated reports should be based on the supplied CV and job description.
- AI results must be returned in a consistent structure that the application can validate.
- Invalid or incomplete AI responses must not be displayed as successful analyses.
- Users should receive understandable feedback when processing fails.
- Previously generated reports should remain accessible without requiring AI to run again.
- CVs and analysis results must remain private to their owners.
- The solution should be achievable using free APIs, open-source models, or suitable free-tier services.

Students may use suitable Node.js libraries for PDF text extraction and an appropriate AI model for qualification analysis. Training a custom AI model is not required.

The main technical challenge is implementing a reliable workflow that connects document processing, AI analysis, response validation, and report presentation.

5. Technology Stack and Deployment

5.1 Required Technology Stack (MERN)

The project must be built exclusively with the MERN stack. No other framework, language, or database is required or expected.

Component	Technology
Database	MongoDB (with Mongoose as the ODM)
Backend	Node.js with Express.js
Frontend	React.js (functional components and hooks, with React Router for navigation)
AI Integration	AI API or suitable model integrated through Express.js
Document Processing	PDF text extraction using an appropriate Node.js solution
Version Control	Git and GitHub

MongoDB is well suited to CareerTech, especially because of its flexible document structures for AI reports and job information. Users, job opportunities, application statuses, CV metadata, and analysis history can be stored in separate collections linked by references.

The Express.js backend is responsible for authentication, authorization, application management, CV processing, AI integration, and database operations. The React frontend consumes the backend REST API.

Students are responsible for designing the database, selecting suitable file storage solutions (for example, MongoDB GridFS or a cloud storage service), organizing the application architecture, and choosing the AI integration approach.

The project does not require a separate AI microservice.

5.2 Deployment

The completed CareerTech application must be deployed and accessible online. The frontend, Express.js backend, database, and AI integration must communicate correctly in the deployed environment.

Suggested deployment services include:

Component	Suggested Service
React Frontend	Vercel or Netlify
Express.js Backend	Railway or Render
MongoDB Database	MongoDB Atlas

These providers are suggestions rather than mandatory choices. Students may select alternative hosting solutions that satisfy the project requirements, preferably using free tiers or available free resources.

The deployed application must support CV uploads, job application tracking, AI analysis, and saved analysis history.

6. Expected Deliverables

The final project submission should include:

- A functional full-stack CareerTech application built with the MERN stack.
- Secure user registration, login, and authentication.
- Private user accounts and appropriate access restrictions.
- Main CV upload, viewing, and replacement.
- Temporary CV uploads for individual AI analyses.
- Job opportunity creation, viewing, editing, and deletion.
- Application status tracking.
- A personal dashboard with real application statistics.
- A dedicated AI CV analysis feature.

- Structured qualification matching and CV improvement reports.
- Saved AI analysis history.
- Persistent database integration using MongoDB.
- Validation and error handling for uploaded documents and AI responses.
- Source code available in a GitHub repository.
- A README containing the project overview, installation steps, configuration, AI integration, and deployment instructions.
- Sample job opportunities, fictional CVs, and analysis results for demonstration.
- A live deployed application accessible online.
- A final presentation demonstrating the application's main workflows and technical decisions.

Example End-to-End Scenario

- **1. Account and CV Setup:** Sara, a junior developer seeking her first full-time role, creates a CareerTech account and uploads her main CV highlighting React, Next.js, TypeScript, and Express.js experience.
- **2. Opportunity Tracking:** She finds a Frontend Developer opening, saves the job title, company, description, and original link, and marks the opportunity as Saved because she has not yet applied.
- **3. CV Analysis Request:** Sara opens the dedicated CV Analysis page, chooses her main CV, and pastes the job description, which asks for React, Next.js, TypeScript, and REST API experience.
- **4. AI Assessment:** CareerTech extracts the CV text and sends it with the job description through the Express.js backend. The AI highlights qualifications supported by the CV and identifies requirements that are not clearly demonstrated.
- **5. Practical CV Improvements:** The report recognizes Sara's projects involving REST APIs and suggests making her API integration work and individual contributions more visible, without inventing experience or qualifications.
- **6. Application Submission:** After reviewing the report and improving her CV, Sara applies through the company's official website and manually changes the opportunity's status from Saved to Applied.
- **7. Interview Tracking:** A week later, she receives an interview invitation, updates the status to Interview, and adds the interview date to her notes.
- **8. Dashboard and Report History:** Sara reviews her application progress on the dashboard and reopens the original AI report without generating it again.
- **9. Alternative CV Analysis:** For a separate Backend Developer position, Sara uploads a backend-focused CV for one analysis. Her main CV remains unchanged, and the new report is saved independently.

This scenario demonstrates how CV management, application tracking, AI-assisted assessment, personalized suggestions, dashboards, and preserved report history work together without automatically applying to jobs or altering application statuses.

7. Evaluation Criteria

The project will be evaluated according to the following criteria:

Criterion	Description
Functional Completeness	Correct implementation of authentication, CV management, job tracking, dashboard statistics, and AI analysis.
Frontend and User Experience	Interface quality, responsiveness, ease of use, and clarity of application management and AI reports.
Backend and Database	Quality of API development, authentication, authorization, data management, and file handling.
AI Integration	Relevance, reliability, and practical usefulness of CV-to-job analysis and improvement suggestions.
AI Response Quality	Accuracy, evidence grounding, structured output validation, and handling of insufficient information.
Code Quality	Organization, readability, maintainability, and appropriate technical decisions.
Deployment	Accessibility and correct integration of the frontend, backend, database, and AI functionality.
Technical Understanding	Ability to explain the application's workflows, AI integration, and implementation decisions.
Final Demonstration	Ability to demonstrate the complete application and its main workflows.

The expected outcome is a practical, complete, and accessible personal job-search platform that demonstrates the ability to build a full-stack MERN application, manage private user data, process PDF documents, and integrate artificial intelligence to help job seekers understand and improve how their qualifications are presented for specific opportunities.